MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3830 REAGENTS - FLOTATION REAGENTS EQP
FDN 3830-REA-001,002 & 3830-TNK-001,002,003

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
FDN.	:	Foundation
EQP	:	Equipment

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1. INTENT

This document is to prepare design of foundation for **3830-REA-001,002 & 3830-TNK-001,002 & 003**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Equipment load data drawings & designed in excel sheet.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Input drawings attached in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Wall	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.2 m below FGL. (Refer Table 5 - BH-10 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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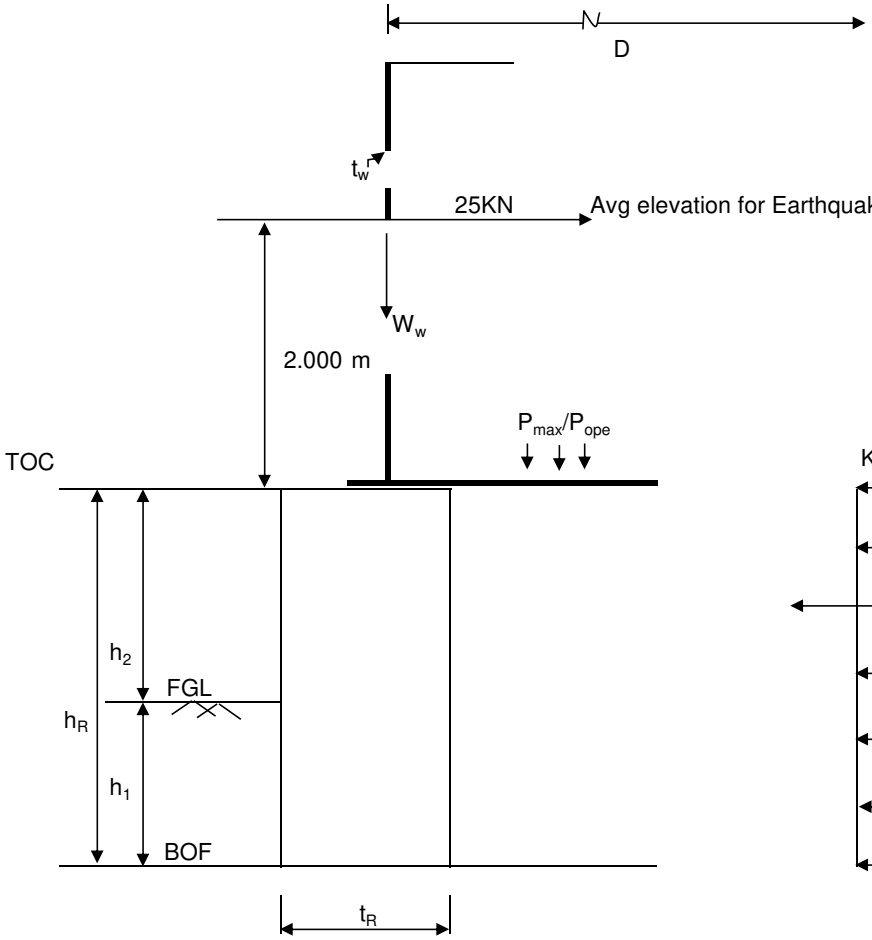
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6. DESIGN CALCULATION FOR FOUNDATION

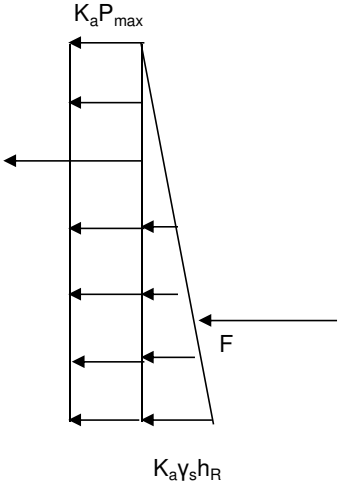
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Design Calculation of Foundation for 3830-TNK-001 & 003



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	3.50	m
Self weight of tank =	40.0	KN
Self weight of tank per m of circumference, W_w =	3.64	KN/m
Design height of liquid, h_L =	3.70	m
Maximum weight of Fluid=	380	KN
Pressure at base of tank , P_{max}	39.5	KN/m^2
Operating weight of Fluid=	330	KN
Pressure at base of tank , P_{ope}	34.3	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete, γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil, γ_s =	20	Kn/m^3
Allowable bearing pressure, q_{all}=	350	Kn/m^2
Coefficient of active earth pressure, K_a =	0.33	

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.05	m
FGL to Bottom of foundation , h_1 =	1.2	m
FGL to Top of concrete , h_2 =	0.85	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall, D_i =	3.0	m
Outer dia of ringwall, D_o =	3.9	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	3.78	m^3

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Check of soil bearing pressure (Operating Fluid Load): For Earthquake Load

Earthquake force = 25 kn acting at a height of 4.050 m from BOF

M_{OT} = Overturning moment = 101.3 KN-m

M_{OT} / S_{xx} = 26.8 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{ope}/2 + h_R\gamma_c$ = 74.4 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 93.2 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 55.7 KN/m^2

Check of soil bearing pressure (Empty Tank): For Earthquake Load

Soil bearing pressure under ringwall :

Load Combination : 0.6 Dead +/- 0.7 Earthquake load

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 57.3 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 53.1 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 15.64 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For Earthquake Load

M_{OT} = Overturning moment = 101.3 KN-m

M_{OR} = Restoring moment = (40+PI()*3.5*0.45*2.05*24)*3.5/2 = 496.0 KN-m

FOS = M_{OR} / M_{OT} 4.90 > 1.5 ok

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.6P_{max} + h_R\gamma_c$ = 81.0 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 80.5 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 250mm out of total 450mm , the pressure under the tank is reduced to 250/450 = 0.6 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 40.6 Kn/m

For Hydrotest :

Radial outward force on ringwall, $F = K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 38.9 Kn/m

Hoop tension in ringwall, T = 1/2FD = 71 Kn

Factored design tension, T_u= 99.4 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 263 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1845 mm^2

$A_{s,reqd}$ (per face of ring wall)= 923 mm^2

Required : 12 dia bar @ 252 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1546 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

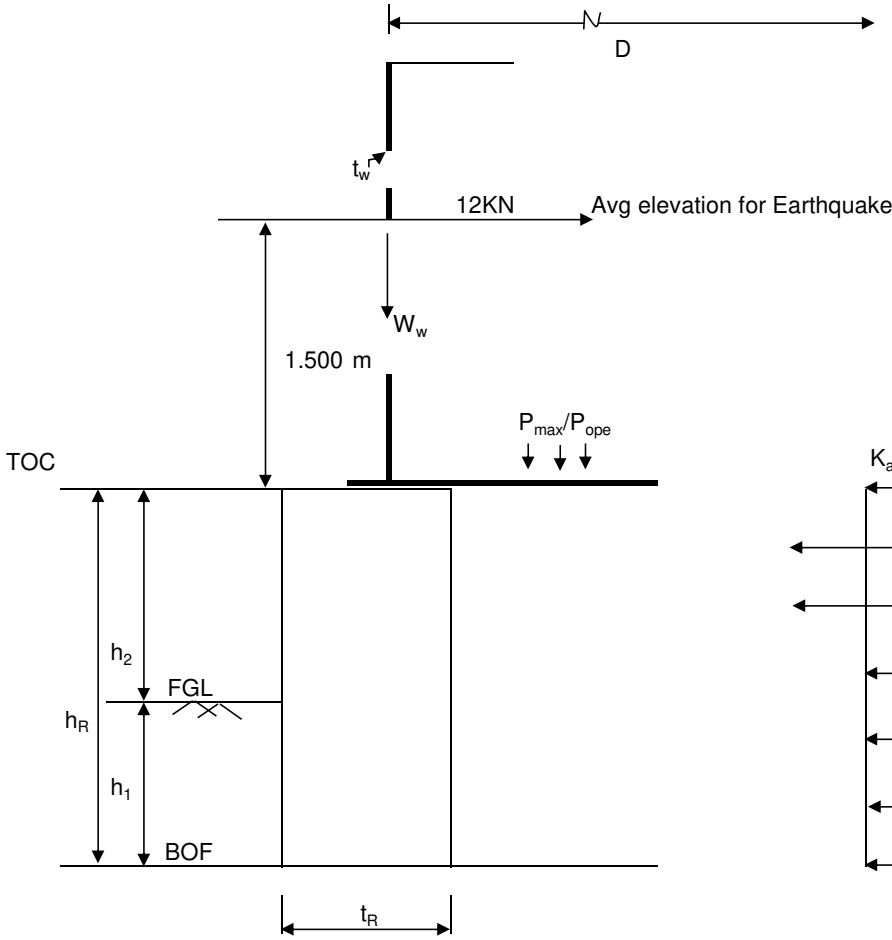
$A_{s,\ provided}$(per face) = 393 mm^2 ok

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Design Calculation of Foundation for 3830-REA-001 & 002



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	2.37	m
Self weight of tank =	36.0	KN
Self weight of tank per m of circumference,W_w =	4.84	KN/m
Design height of liquid, h_L =	2.70	m
Maximum weight of Fluid=	180	KN
Pressure at base of tank ,P_{max}	40.8	KN/m^2
Operating weight of Fluid=	130	KN
Pressure at base of tank ,P_{ope}	29.5	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	20	Kn/m^3
Allowable bearing pressure, q_{all}=	350	Kn/m^2
Coefficient of active earth pressure,K_a =	0.33	

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.05	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	0.85	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall,D_i =	1.92	m
Outer dia of ringwall,D_o =	2.82	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	1.73	m^3

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Check of soil bearing pressure (Operating Fluid Load): For Earthquake Load

Earthquake force = 12 kn acting at a height of 3.550 m from BOF

M_{OT} = Overturning moment = 42.6 KN-m

M_{OT} / S_{xx} = 24.6 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{ope}/2 + h_R\gamma_c$ = 74.7 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 91.9 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 57.4 KN/m^2

Check of soil bearing pressure (Empty Tank): For Earthquake Load

Soil bearing pressure under ringwall :

Load Combination : 0.6 Dead +/- 0.7 Earthquake load

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 59.9 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 53.2 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 18.72 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For Earthquake Load

M_{OT} = Overturning moment = 42.6 KN-m

M_{OR} = Restoring moment = (36+PI()*2.37*0.45*2.05*24)*2.37/2 = 238.0 KN-m

FOS = M_{OR} / M_{OT} 5.59 > 1.5 ok

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.5P_{max} + h_R\gamma_c$ = 80.3 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 81.8 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 225mm out of total 450mm , the pressure under the tank is reduced to 225/450 = 0.5 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 41.5 Kn/m

For Hydrotest :

Radial outward force on ringwall, $F = K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 32.1 Kn/m

Hoop tension in ringwall, T = 1/2FD = 49.1 Kn

Factored design tension, T_u= 68.8 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 182 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1845 mm^2

$A_{s,reqd}$ (per face of ring wall)= 923 mm^2

Required : 12 dia bar @ 252 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1546 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

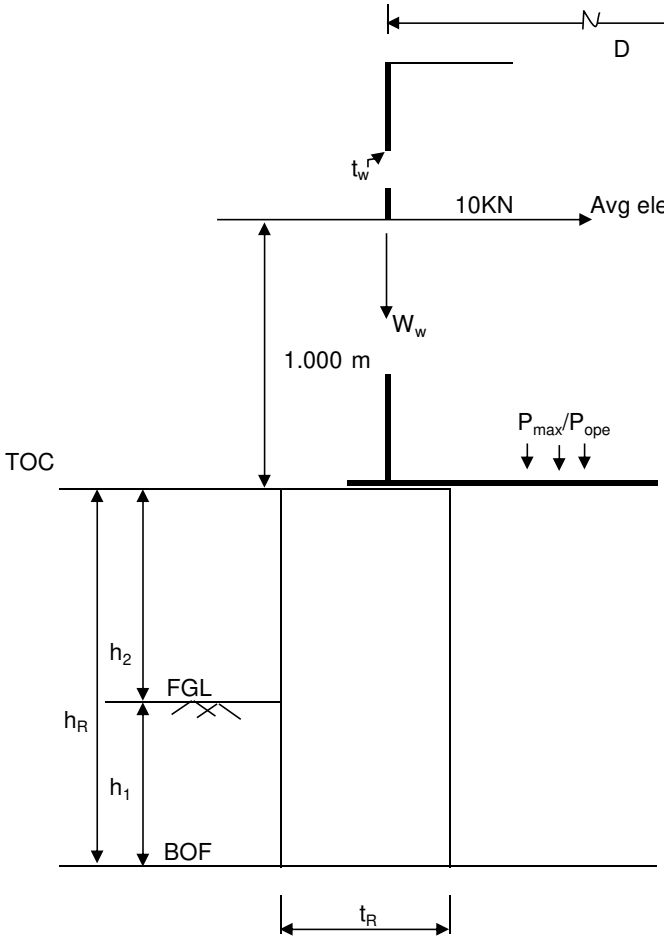
$A_{s,\ provided}$(per face) = 393 mm^2 ok

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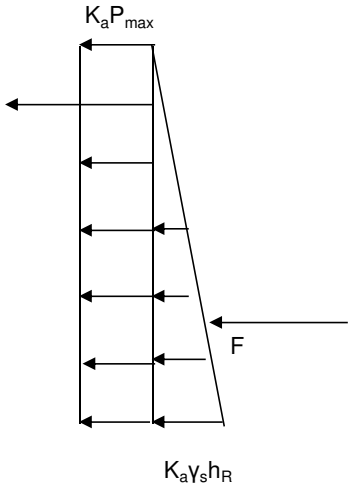
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Design Calculation of Foundation for 3830-TNK-002



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	2.00	m
Self weight of tank =	30.0	KN
Self weight of tank per m of circumference, W_w =	4.77	KN/m
Design height of liquid, h_L =	1.70	m
Maximum weight of Fluid=	70	KN
Pressure at base of tank ,P_{max}	22.3	KN/m^2
Operating weight of Fluid=	58	KN
Pressure at base of tank ,P_{ope}	18.5	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	20	Kn/m^3
Allowable bearing pressure, q_{all}=	350	Kn/m^2
Coefficient of active earth pressure,K_a =	0.33	

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.05	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	0.85	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall,D_i =	1.8	m
Outer dia of ringwall,D_o =	2.7	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	1.55	m^3

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Check of soil bearing pressure (Operating Fluid Load): For Earthquake Load

Earthquake force = 10 kn acting at a height of 3.050 m from BOF

M_{OT} = Overturning moment = 30.5 KN-m

M_{OT} / S_{xx} = 19.7 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{ope}/2 + h_R\gamma_c$ = 69 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 82.8 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 55.3 KN/m^2

Check of soil bearing pressure (Empty Tank): For Earthquake Load

Soil bearing pressure under ringwall :

Load Combination : 0.6 Dead +/- 0.7 Earthquake load

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 59.8 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 49.7 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 22.12 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For Earthquake Load

M_{OT} = Overturning moment = 30.5 KN-m

M_{OR} = Restoring moment = (30+PI()*2*0.45*2.05*24)*2/2 = 169.1 KN-m

FOS = M_{OR} / M_{OT} 5.54 > 1.5 ok

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.2P_{max} + h_R\gamma_c$ = 64.3 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 63.3 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 100mm out of total 450mm , the pressure under the tank is reduced to 100/450 = 0.2 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 28.9 Kn/m

For Hydrotest :

Radial outward force on ringwall, $F = K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 25.4 Kn/m

Hoop tension in ringwall, T = 1/2FD = 28.9 Kn

Factored design tension, T_u = 40.5 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 107 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1845 mm^2

$A_{s,reqd}$ (per face of ring wall)= 923 mm^2

Required : 12 dia bar @ 252 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1546 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 393 mm^2 ok

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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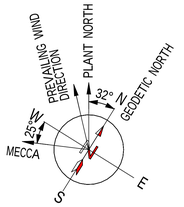
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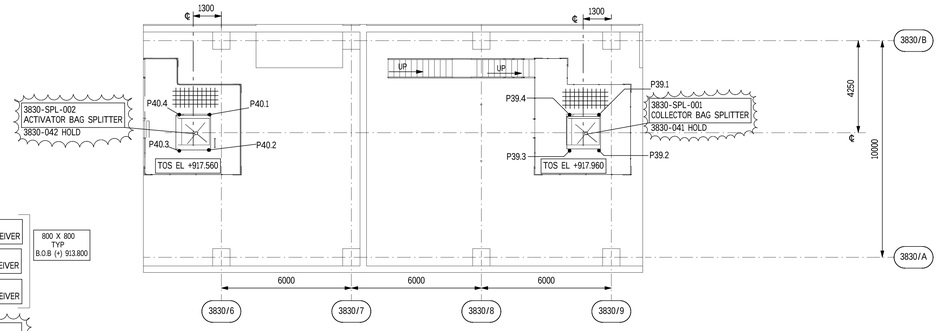
ATTACHMENT 1 INPUT DRAWINGS

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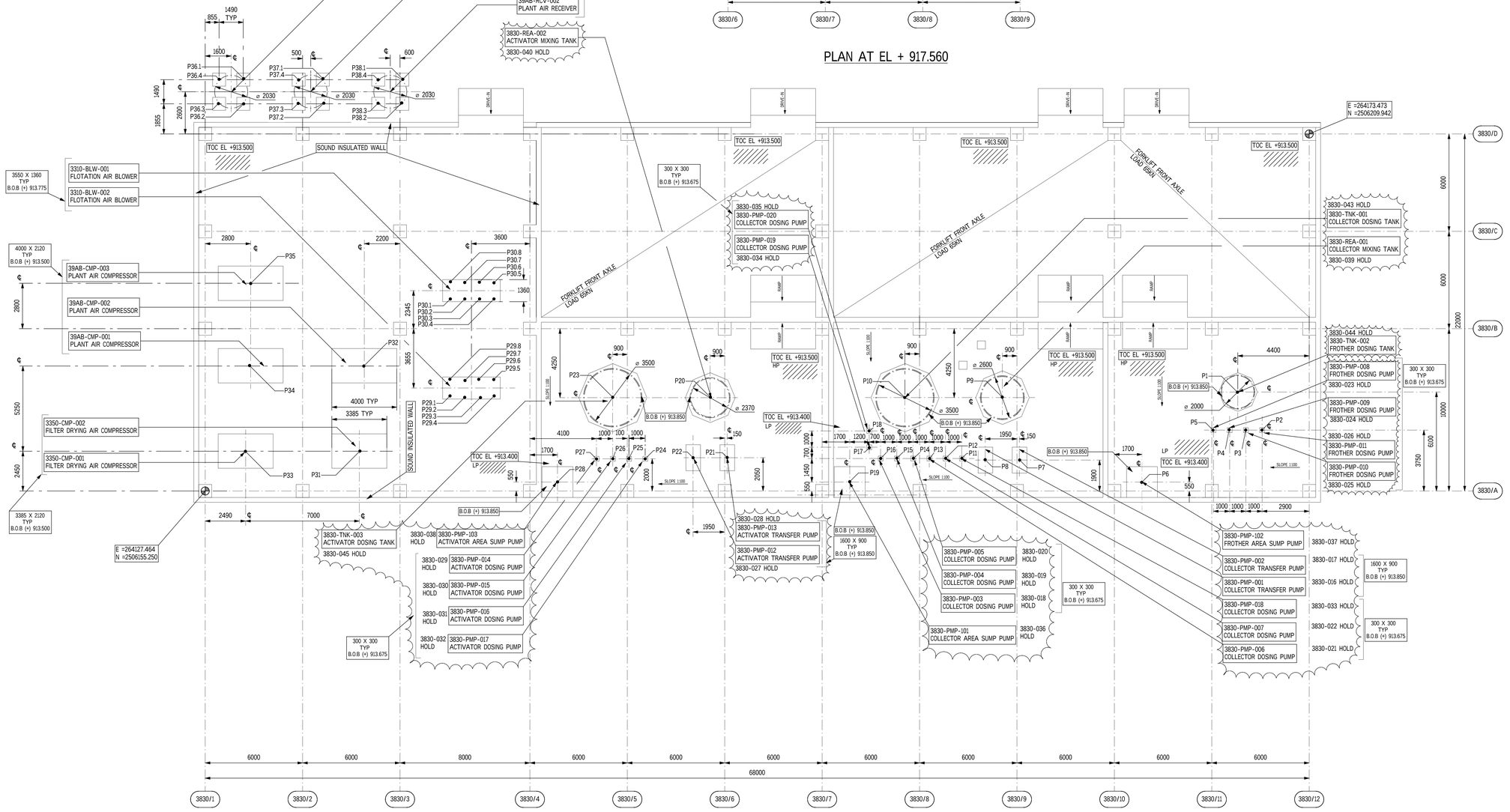
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DRAWING NUMBER: MD-216-3830-EG-ME-GAD-1017 | REV: A01





PLAN AT EL + 917.560



PLAN AT EL +913.000

0 1 10m

	CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
	CONCRETE FLOOR
A	CONCRETE FLOOR WITH ACID RESISTANT COATING
	CHECKERED PLATE
	GRATING
F	GRATING - FRP
	ACID RESISTANT BRICKLINING
	BRICK WALL
	CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS: +Z +Mz +Y +X +My +Mx

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT.
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

OPERATING PLATFORM- GRATING 5 kN/m²

GROUND FLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²

HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

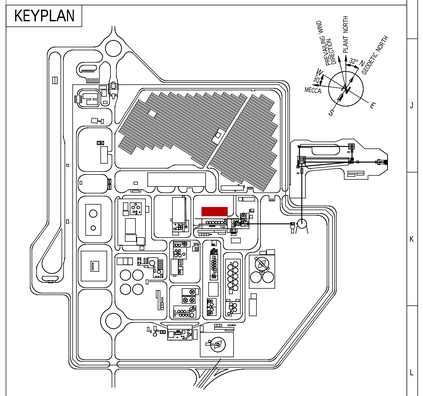
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



Document Number MD-216-3830-EG-ME-GAD-1017
Revision A01
Approved
Mr Doron Golan - Bechtel Global
Feb 19, 2020, 3:11 PM GMT+2:00
Permission to proceed does not constitute acceptance or approval of design detail, calculations, analysis, test methods, or materials developed or selected by the Supplier, and does not relieve the Supplier from full compliance with contractual obligations

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3830-EG-ME-GAD-1019	LOAD TABLE
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3310-EG-ME-GAD-1003	GENERAL ARRANGEMENT DRAWING, PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	24-01-2020	ISSUED FOR APPROVAL	SS	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_D_03279_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AA02_ MLD14 _38301 — DOCUMENT ID: OU100667840 — REVISION: 0

DRAWN	SS	24-01-2020
CHECKED	TV	24-01-2020
ENGINEER	SC	24-01-2020
APPROVED	RL	24-01-2020
EMPLOYER		
SCALE	1:100	A0

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
FLOTATION REAGENTS, BLOWERS AND PLANT AIR
PLAN VIEW

DRAWING NUMBER: MD-216-3830-EG-ME-GAD-1017 — REV: A01

DRAWING NUMBER: MD-216-3830-EG-ME-GAD-1019 REV A01

LOADING POINT	P1				P2				P3				P4				P5				P6				P7				P8				P9				P10				P11				P12				P13				P14				P15			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	-	30			-	3			-	3			-	3			-	3			-	7			-	4			-	4			-	36			-	40			-	3			-	3			-	3			-	3			-	3		
2 DEAD LOAD (SECONDARY EQUIPMENT)																																																								
3 OPERATION LOAD 1. FILLING	-	58																															-	130			-	330																				
4 OPERATION LOAD 2. PRESSURE																																																							
5 OPERATION LOAD 3. TEMPERATURE																																																						
6 OPERATION LOAD 4. DYNAMIC					-	5			-	5			-	5			-	5			-	10			-	6			-	6											-	5			-	5			-	5			-	5			-	5		
7 OPERATION LOAD 5. OTHERS																																																							
8 TEST LOAD FILLING / EMERGENCY																																																						
9 LIVE LOAD	-	70																															-	180			-	380																				
10 WIND ±X - DIRECTION			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-		
11 WIND ±Y - DIRECTION																																																							
12 EARTHQUAKE ±X - DIRECTION			±	10			±	1			±	1			±	1			±	1			±	1			±	1			±	12			±	25			±	1			±	1			±	1			±	1			±	1
13 EARTHQUAKE ±Y - DIRECTION																																																						
EQUIPMENT - NO.	3830-TNK-002				3830-PMP-011				3830-PMP-010				3830-PMP-009				3830-PMP-008				3830-PMP-102				3830-PMP-002				3830-PMP-001				3830-REA-001				3830-TNK-001				3830-PMP-018				3830-PMP-007				3830-PMP-006				3830-PMP-005				3830-PMP-004			
DESCRIPTION	FROTHER				FROTHER				FROTHER				FROTHER				FROTHER				FROTHER AREA				COLLECTOR				COLLECTOR				COLLECTOR				COLLECTOR				COLLECTOR				COLLECTOR				COLLECTOR				COLLECTOR			
	DOSING TANK				DOSING PUMP				DOSING PUMP				DOSING PUMP				DOSING PUMP				SUMP PUMP				TRANSFER PUMP				TRANSFER PUMP				MIXING TANK				DOSING TANK				DOSING PUMP				DOSING PUMP				DOSING PUMP				DOSING PUMP			

LOADING POINT	P16				P17				P18				P19				P20				P21				P22				P23				P24				P25				P26				P27				P28				P29.1				P29.2			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	-	3			-	3			-	3			-	7			-	36			-	4			-	4			-	40			-	3			-	3			-	3			-	3			-	3			-	6			-	4		
2 DEAD LOAD (SECONDARY EQUIPMENT)																																																								
3 OPERATION LOAD 1. FILLING																	-	130											-	330																												
4 OPERATION LOAD 2. PRESSURE																																																							
5 OPERATION LOAD 3. TEMPERATURE																																																						
6 OPERATION LOAD 4. DYNAMIC	-	5			-	5			-	5			-	10							-	6			-	6							-	5			-	5			-	5			-	5			-	6			-	4		
7 OPERATION LOAD 5. OTHERS																																																							
8 TEST LOAD FILLING / EMERGENCY																																																						
9 LIVE LOAD																													-	380																												
10 WIND ±X - DIRECTION			-	-			-	-			-	-			-	-	-	180	-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-		
11 WIND ±Y - DIRECTION																																																							
12 EARTHQUAKE ±X - DIRECTION			±	1			±	1			±	1			±	1			±	12			±	1			±	1			±	25			±	1			±	1			±	1			±	1			±	1			±	1
13 EARTHQUAKE ±Y - DIRECTION																																																						
EQUIPMENT - NO.	3830-PMP-003				3830-PMP-019				3830-PMP-020				3830-PMP-101				3830-REA-002				3830-PMP-012				3830-PMP-013				3830-TNK-003				3830-PMP-017				3830-PMP-016				3830-PMP-015				3830-PMP-014				3830-PMP-103				3310-BLW-002				3310-BLW-002			
DESCRIPTION	COLLECTOR				COLLECTOR				COLLECTOR				COLLECTOR AREA				ACTIVATOR				ACTIVATOR				ACTIVATOR				ACTIVATOR				ACTIVATOR				ACTIVATOR				ACTIVATOR				ACTIVATOR AREA				FLOTATION AIR				FLOTATION AIR			
	DOSING PUMP				DOSING PUMP				DOSING PUMP				SUMP PUMP				MIXING TANK				TRANSFER PUMP				TRANSFER PUMP				DOSING TANK				DOSING PUMP				DOSING PUMP				DOSING PUMP				DOSING PUMP				SUMP PUMP				BLOWER				BLOWER			

LOADING POINT	P29.3				P29.4				P29.5				P29.6				P29.7				P29.8				P30.1				P30.2				P30.3				P30.4				P30.5				P30.6				P30.7				P30.8			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	-	4.5			-	5			-	5.5			-	4.5			-	4			-	6			-	6			-	4			-	4.5			-	5			-	5.5			-	4.5			-	4			-	6		
2 DEAD LOAD (SECONDARY EQUIPMENT)																																																								
3 OPERATION LOAD 1. FILLING																																																						
4 OPERATION LOAD 2. PRESSURE																																																						
5 OPERATION LOAD 3. TEMPERATURE																																																						
6 OPERATION LOAD 4. DYNAMIC	-	4			-	5			-	5.5			-	4.5			-	4			-	6			-	5.5			-	4			-	4			-	5			-	5.5			-	4.5			-	4			-	6		
7 OPERATION LOAD 5. OTHERS																																																						
8 TEST LOAD FILLING / EMERGENCY																																																					
9 LIVE LOAD																																																					
10 WIND ±X - DIRECTION			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-			-	-		
11 WIND ±Y - DIRECTION																																																					
12 EARTHQUAKE ±X - DIRECTION			±	1			±	1			±	1			±	1			±	1			±	1			±	1			±	1			±	1			±	1			±	1			±	1			±	1			±	1
13 EARTHQUAKE ±Y - DIRECTION																																																					
EQUIPMENT - NO.	3310-BLW-002				3310-BLW-002				3310-BLW-002				3310-BLW-002				3310-BLW-002				3310-BLW-002				3310-BLW-001				3310-BLW-001				3310-BLW-001				3310-BLW-001				3310-BLW-001				3310-BLW-001				3310-BLW-001				3310-BLW-001			
DESCRIPTION	FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR				FLOTATION AIR			
	BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER				BLOWER			

LOADING POINT	P31				P32				P33				P34				P35				P36.1/..../36.4				P37.1/...../37.4				P38.1/...../38.4				P39.1/..../39.4				P40.1/...../40.4			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	-	55			-	80			-	55			-	80			-	80			-	45			-	45			-	45			-	4			-	4		
2 DEAD LOAD (SECONDARY EQUIPMENT)																																								
3 OPERATION LOAD 1. FILLING																																	-	6			-	6		
4 OPERATION LOAD 2. PRESSURE																																								
5 OPERATION LOAD 3. TEMPERATURE																																								
6 OPERATION LOAD 4. DYNAMIC																																								
7 OPERATION LOAD 5. OTHERS																																								
8 TEST LOAD FILLING / EMERGENCY																																								
9 LIVE LOAD	-	55			-	80			-	55			-	80			-	80			-	45			-	45			-	45			-	4			-	4		
10 WIND ±X - DIRECTION			-	-			-	-			-	-			-	-			-	-	±	11.4	±	5.7	±	5.7	±	11.4	±	5.7	±	5.7	±	11.4	±	5.7	±	5.7
11 WIND ±Y - DIRECTION																					±	11.4	±	5.7	±	5.7	±	11.4	±	5.7	±	5.7	±	11.4	±	5.7	±	5.7
12 EARTHQUAKE ±X - DIRECTION			±	6			±	8			±	6			±	8			±	8			±	5			±	5			±	5			±	1			±	1
13 EARTHQUAKE ±Y - DIRECTION																																								
EQUIPMENT - NO.	3350-CMP-002				39AB-CMP-002				3350-CMP-001				39AB-CMP-001				39AB-CMP-003				3350-RCV-101				39AB-RCV-001				39AB-RCV-002				3830-SPL-001				3830-SPL-002			
DESCRIPTION	FILTER DRYING				PLANT AIR				FILTER DRYING				PLANT AIR				PLANT AIR				FILTER DRYING				PLANT AIR				PLANT AIR				COLLECTOR BAG				ACTIVATOR BAG			
	AIR COMPRESSOR				COMPRESSOR				AIR COMPRESSOR				COMPRESSOR				COMPRESSOR				AIR RECEIVER				RECEIVER				RECEIVER				SPLITTER				SPLITTER			

Please define to which code these loads have been calculated so that foundation can be designed.

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTON METER[kNm] OR KILONEWTON/SQUARE METER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS



06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

OPERATING PLATFORM- GRATING 5 kN/m²

GROUND FLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²

HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

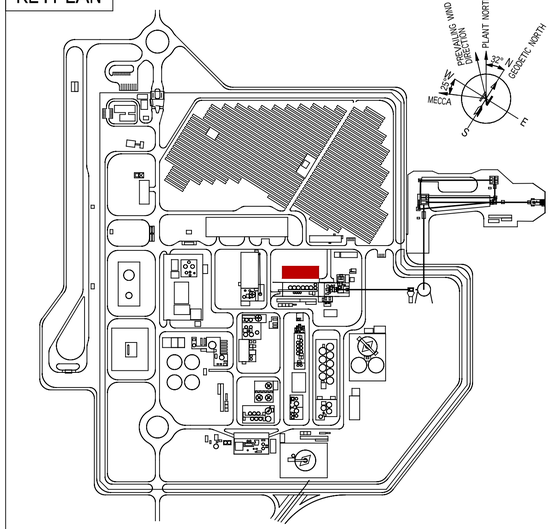
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



BECHTEL

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Revision A01

Approved with Comments

Mr Doron Golan - Bechtel Global

Feb 19, 2020, 4:55 PM GMT+2:00

Permission to proceed does not constitute acceptance or approval of design detail, calculations, analysis, test methods, or materials developed or selected by the Supplier, and does not relieve the Supplier from full compliance with contractual obligations

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP	DATE	REV	REFERENCE DOCUMENTS
ISSUED FOR APPROVAL	SS	TV	SC	RL		24-01-2020	A01	LOAD PLAN, PLAN VIEW
								HOLD LIST

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER | REVISION 0

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CHECKED	TV	24-01-2020
ENGINEER	SC	24-01-2020
APPROVED	RL	24-01-2020
EMPLOYER		

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
FLOTATION REAGENTS, BLOWERS AND PLANT AIR LOAD TABLE

SCALE 1:100 A1

DRAWING NUMBER MD-216-3830-EG-ME-GAD-1019 | REV A01